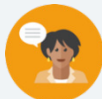


7.8 Factoring the Difference of Two Squares

Lesson Introduction				
 Benchmark	MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.		 Learning Targets	- I can factor a difference of two squares. - I can rewrite a polynomial expression as a product of polynomials by factoring.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.		MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system.	
 Essential Questions	- How can a polynomial with two terms be factored? - How can you recognize and factor a difference of two squares?			
 Lesson Narrative	In this lesson, students conclude their journey with factoring by learning about the final special case included in this unit, a difference of two squares. Students begin by exploring a difference of squares using algebra tiles. They will then practice identifying whether a given expression is a difference of squares. Students will then learn how to factor a situation represented by a difference of squares. The lesson includes components of mixed-review practice of factoring given a variety of polynomials.			
 Component Summary	7.8.1 Warm-Up (~5 min.)	Factor perfect square trinomials (<i>SE p. 398</i>)	7.8.4 Try It (10 min.)	Mixed practice factoring polynomials (<i>SE p. 401</i>)
	7.8.2 Exploration (15 min.)	Exploration activity to discover a difference of two squares (<i>SE p. 398</i>)	7.8.5 Your Turn! (5-10 min.)	Independent practice factoring polynomials (<i>SE p. 402</i>)
	7.8.3 Guided Practice (10 min.)	Guided practice with factoring difference of two squares (<i>SE p. 400</i>)	7.8.6 Wrap-Up (~5 min.)	Self-reflection on student understanding of factoring from this unit (<i>SE p. 402</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Difference of two squares <u>Additional Terms:</u> N/A		N/A	

 Teacher Tips	Common Misconceptions • Students may mistake the difference of squares for a perfect square trinomial. • Students may not recognize a difference of squares where the leading coefficient is negative but the sign between the terms is addition.	Things to Consider • A difference of squares was shown pictorially in Lesson 7 but not formally introduced. This lesson provides not only instruction on difference of squares but also a mixed review of factoring all polynomial types explored in this unit. • If time is an issue, consider assigning 7.8.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Poll the Class The table component in 7.8.2 Exploration provides an opportunity for students to determine if a series of conditions are met for a given polynomial. Consider leveraging this protocol when reviewing the answers to the table to determine who in the class thinks it meets a given condition and who does not.	MA.K12.MTR.3.1: Mathematical Fluency 7.8.4 Try It provides students with a mixed-review opportunity for factoring polynomials using the methods learned throughout this unit. Students will select efficient and appropriate methods for factoring given expressions. They should maintain flexibility and accuracy when factoring different types of polynomials. The lesson offers multiple opportunities for students to practice efficient and generalizable methods.
 Differentiate Instruction	When reviewing 7.8.2 Exploration, have students add the formal definition, with examples and nonexamples, to their current factoring flip charts, Frayer Models, or guided notes that they have for this unit. For 7.8.4 Try It, consider grouping students together based on strengths from this unit. This activity is a mixed review, and having students grouped by strengths from the unit can help them learn better from one another.	
Lesson Notes:		

7.8.1 Warm-Up: Bell Work

Component Overview: Students will complete two factoring problems that are both perfect square trinomials.



7.8.1 Warm-Up: Bell Work

1. Factor $36x^2 - 12x + 1$.
 $(6x - 1)^2$
2. Factor $9y^2 + 30y + 25$.
 $(3y + 5)^2$



Use inquiry to extend student thinking from Lesson 7. Is this a perfect square trinomial? How do you know? How did that impact your method for factoring?

7.8.2 Exploration: What Is a Difference of Two Squares?

Component Overview: Students will work together to explore the concept of a difference of two squares. They will use algebra tiles to conceptually develop understanding of this concept and relate it to what they've learned in this unit.



7.8.2 Exploration: What is a Difference of Two Squares?

The table shows two algebra tile models and their corresponding expressions.

$4x^2 - 12x + 9$	$4x^2 - 9$

The quadratic expression $4x^2 - 12x + 9$ is a perfect square trinomial.

- Discuss with your partner why the quadratic expression $4x^2 - 9$ is not a perfect square trinomial, even though the algebra tile model is a square. Summarize your discussion.

There is no middle term to meet the conditions of a perfect square trinomial.

- Use the algebra tile models to factor each quadratic expression.

$4x^2 - 12x + 9$	$4x^2 - 9$
$(2x - 3)^2$ or $(2x - 3)(2x - 3)$	$(2x + 3)(2x - 3)$

- What do you notice about the two factored forms?
They are similar in terms, but the second cannot be written as a square since there are two binomials with different signs. Or: In the first, the signs are the same for both binomials, and in the second, the signs are different.



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 7.8.3 Guided Practice.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



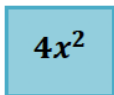
Monitor for student responses to revisit or highlight during whole group synthesis. For example, students may identify the similarities of using square roots while also highlighting the difference in a missing b term. Prompt this response whole group later as a launch for synthesizing the difference between a perfect square trinomial and a difference of two squares.

7.8.2 Exploration: What Is a Difference of Two Squares? (continued)

4. The quadratic expression $4x^2 - 9$ is a difference of two squares. Discuss with your partner why $4x^2 - 9$ is called a "difference."

Answers will vary. The difference refers to the subtraction.

5. The two squares shown represent each term of the binomial $4x^2 - 9$.



What is the side length of each square?

$4x^2$	9
Side Length = $2x$	Side Length = 3

The squares are a geometric representation of each term. The quadratic expression $4x^2 - 9$ can be rewritten with each term as a monomial squared, as shown.

$$(2x)^2 - (3)^2$$

6. Discuss with your partner how the factored form of $4x^2 - 9$ reflects the expression $(2x)^2 - (3)^2$. Summarize your discussion.

Answers will vary. The first term of each binomial is $2x$ and the second term is 3.



The **difference of two squares** is a binomial of the form $(a)^2 - (b)^2$, where a and b can represent a monomial expression or any value except 0. The factored form is $(a + b)(a - b)$.



Provide an opportunity for students to share their answers to questions 5 and 6.



Have students add the formal definition of the difference of two squares into their existing guided notes, anchor charts, flip charts, foldables, Frayer Models, or other reference documents created for this unit.

7.8.2 Exploration: What Is a Difference of Two Squares? (continued)

7. Use the table to determine if each expression is a difference of two squares.

Quadratic Expression	The Expression is a Binomial that Can Be Written as a Difference	First Term is a Perfect Square	Second Term is a Perfect Square
$-100g^4 + 1$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$16y^2 - 100y$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$-b^2 + 25$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$25w^2 - 40w + 16$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$9x^2 - 4y^6$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$q^2 - 2qz + z^2$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$49g^2 - 81h^2$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

- Complete the table.
- Circle the expressions in the table that are considered a difference of two squares, based on your check marks.



This table purposely starts with a binomial that is a difference of two squares but is not in the recognizable form. The 1 is also purposely used to ensure that you are able to talk about leading terms that are numeric. This is also seen again in the third row.

Allow students to work through question 7 with their partners, but be sure to review any errors, paying particular attention to these examples, as needed before transitioning to the next component.

7.8.3 Guided Practice: Factoring a Difference of Two Squares

Component Overview: Students will practice factoring the difference of squares with three different polynomials. For this activity, students will need a partner in order to engage in all the questions.



7.8.3 Guided Practice: Factoring a Difference of Two Squares

1. Consider the expression $49g^2 - 81h^2$.
 - a. Rewrite $49g^2 - 81h^2$ as a difference of two squares.
 $(7g)^2 - (9h)^2$
 - b. Factor $49g^2 - 81h^2$.
 $(7g + 9h)(7g - 9h)$

2. Consider the expression $9x^2 - 4y^6$.
 - a. Rewrite $9x^2 - 4y^6$ as a difference of two squares.
 $(3x)^2 - (2y^3)^2$
 - b. Factor $9x^2 - 4y^6$.
 $(3x + 2y^3)(3x - 2y^3)$

3. Both $-121y^2 + 81$ and $-1 + 16x^2$ are differences of two squares. Discuss with your partner how to rewrite them as a difference of two squares. Then, work together to factor each.



Use observational data from the 7.8.2 Exploration to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 to address common misconceptions shown in the 7.8.2 Exploration before releasing students to work on questions 2 and 3 with their partners.

$-121y^2 + 81$	$-1 + 16x^2$
$81 - 121y^2$	$16x^2 - 1$
$(9 - 11y)(9 + 11y)$	$(4x + 1)(4x - 1)$

7.8.4 Try It: Factoring Polynomials

Component Overview: Students will practice factoring using the methods learned throughout the unit.



7.8.4 Try It: Factoring Polynomials

This is a mixed practice, which includes all of the different types of polynomials factored in this unit. If necessary, review previous lessons for examples of how to factor each type.

1. Factor each expression.

a. $49t^2 - 42t + 9$
 $(7t - 3)^2$

b. $1 - 25k^2$
 $(1 + 5k)(1 - 5k)$

c. $36x^2y^7 - 48x^4y^4 + 12x^2y^4$
 $12x^2y^4(3y^3 - 4x^2 + 1)$

d. $12k^2 + 3k - 15$
 $3(k - 1)(4k + 5)$

e. $d^2 + d - 42$
 $(d - 6)(d + 7)$



This activity is not limited to polynomials that can be factored using the difference of squares. Additional methods from prior lessons will be necessary to completely factor each expression.



Consider having students write the methods that they used to factor each of the expressions in this component. Provide students a reference point to specific lessons they can go back to if they get stumped.

7.8.5 Your Turn!

Component Overview: Students will independently practice factoring polynomials using different methods learned throughout the unit.



7.8.5 Your Turn!

1. Factor each expression.

a. $25m^2 + 20m + 4$
 $(5m + 2)^2$

b. $2g^2 + 3g - 20$
 $(2g - 5)(g + 4)$

c. $y^2 - 5y - 84$
 $(y - 12)(y + 7)$

d. $49z^4 - 16w^4$
 $(7z^2 + 4w^2)(7z^2 - 4w^2)$



This activity is not limited to polynomials that can be factored using the difference of squares. Additional methods from prior lessons will be necessary in order to completely factor each one.



Consider having students write the methods that they used to factor each of the questions in this activity. Provide students a reference point to specific lessons they can go back to if they get stumped.

7.8.6 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection on their levels of understanding of each of the factoring methods learned throughout the unit.



7.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.

Answers will vary.



I need help.



I got it.



I could help someone.

Factoring out a common monomial factor			
Factoring a trinomial where $a = 1$			
Factoring a trinomial where $a > 1$			
Factoring a perfect square trinomial			
Factoring a difference of two squares			



What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of these learning targets throughout the unit and how to best address potential gaps in understanding prior to the unit assessment.

Consider making notes on the unit guide for things you would want to keep in mind for this unit next school year.

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